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AIM

Implement and analyze various probability distributions in Python, including Binomial, Poisson, and Normal distributions.

Theory

1.What is the difference between Binomial, Poisson, and Normal distributions?

Binomial Distribution Deals with discrete outcomes (success/failure). Fixed number of trials (n). Each trial has the same probability (p). Example: Tossing a coin 10 times and counting heads.

Poisson Distribution Also discrete, but used for counting events in a fixed interval (time/space). Events occur independently with a known average rate (λ). Example: Number of calls received in a call center per hour.

Normal Distribution Continuous distribution (not discrete). Symmetrical bell-shaped curve. Defined by mean (μ) and standard deviation (σ). Example: Heights of people in a population.

2.What is the real-world significance of the Poisson distribution?

The Poisson distribution is useful when we want to model rare events over time or space. Real-life applications: Number of emails received per hour Number of accidents at a traffic signal Number of defects in a manufacturing process Number of customers arriving at a store.

3.What is the Central Limit Theorem and how does it relate to Normal distribution?

The Central Limit Theorem states: When we take a large number of samples from any distribution, the sampling distribution of the mean will tend to be Normal, regardless of the original distribution. Relation to Normal Distribution: Even if data is not normal, the average of samples becomes normal. This is why Normal distribution is widely used in statistics. Example: Rolling a dice (not normal) But average of many rolls → becomes approximately normal

4.Why is Normal distribution widely used in machine learning and statistics?

The Normal distribution is widely used because: Many real-world data naturally follow it (heights, marks, errors) It is mathematically simple and well-understood Works well with the Central Limit Theorem Many ML algorithms assume normality (e.g., Gaussian Naive Bayes) Helps in: Hypothesis testing Confidence intervals Data modeling In short: It simplifies analysis and works for many real datasets.

5.What is skewness in distributions and which distributions exhibit it?

Skewness means the asymmetry of a distribution.

Types of skewness:

Positive skew (right-skewed) Tail is longer on the right

Example: Income distribution

Negative skew (left-skewed) Tail is longer on the left

Zero skew (symmetrical) Example: Normal distribution

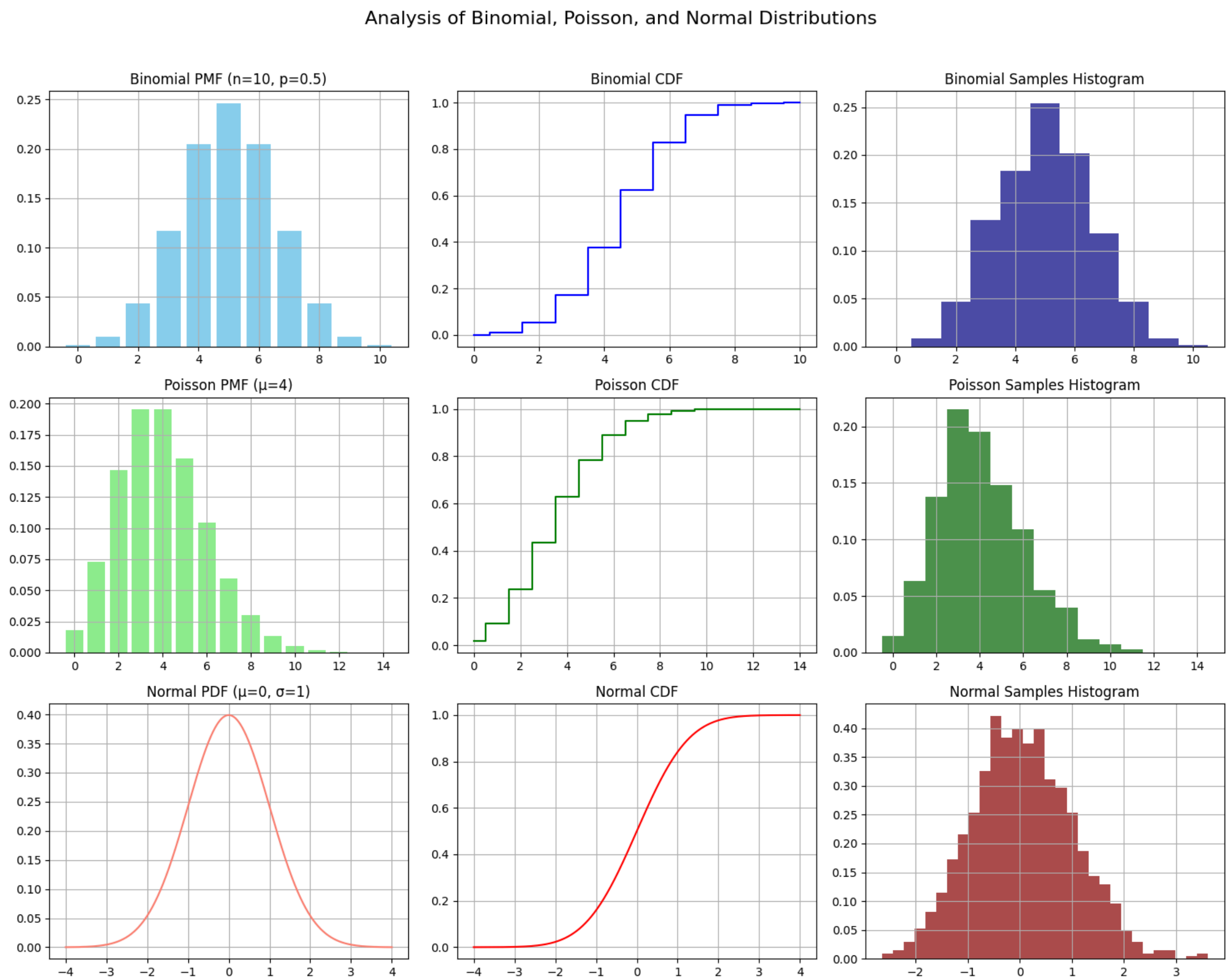
Which distributions show skewness? Binomial Distribution → Can be skewed (depends on p) Poisson Distribution → Usually positively skewed Normal Distribution → Always symmetrical (no skewness)

In [ ]:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, poisson, norm

# Set up a figure with subplots
fig, axs = plt.subplots(3, 3, figsize=(15, 12))
fig.suptitle("Analysis of Binomial, Poisson, and Normal Distributions", fontsi

# ------------------- BINOMIAL DISTRIBUTION -------------------
n, p = 10, 0.5
x_binom = np.arange(0, n+1)
pmf_binom = binom.pmf(x_binom, n, p)
cdf_binom = binom.cdf(x_binom, n, p)
samples_binom = binom.rvs(n, p, size=1000)

# PMF
axs[0, 0].bar(x_binom, pmf_binom, color='skyblue')
axs[0, 0].set_title("Binomial PMF (n=10, p=0.5)")
# CDF
```

```
axs[0, 1].step(x_binom, cdf_binom, where='mid', color='blue')
axs[0, 1].set_title("Binomial CDF")
# Histogram of Samples
axs[0, 2].hist(samples_binom, bins=np.arange(n+2)-0.5, density=True, color='na
axs[0, 2].set_title("Binomial Samples Histogram")

# ------------------- POISSON DISTRIBUTION -------------------
mu = 4
x_poisson = np.arange(0, 15)
pmf_poisson = poisson.pmf(x_poisson, mu)
cdf_poisson = poisson.cdf(x_poisson, mu)
samples_poisson = poisson.rvs(mu, size=1000)

# PMF
axs[1, 0].bar(x_poisson, pmf_poisson, color='lightgreen')
axs[1, 0].set_title("Poisson PMF (μ=4)")
# CDF
axs[1, 1].step(x_poisson, cdf_poisson, where='mid', color='green')
axs[1, 1].set_title("Poisson CDF")
# Histogram of Samples
axs[1, 2].hist(samples_poisson, bins=np.arange(16)-0.5, density=True, color='c
axs[1, 2].set_title("Poisson Samples Histogram")

# ------------------- NORMAL DISTRIBUTION -------------------
mu_norm, sigma = 0, 1
x_norm = np.linspace(-4, 4, 1000)
pdf_norm = norm.pdf(x_norm, mu_norm, sigma)
cdf_norm = norm.cdf(x_norm, mu_norm, sigma)
samples_norm = norm.rvs(mu_norm, sigma, size=1000)

# PDF
axs[2, 0].plot(x_norm, pdf_norm, color='salmon')
axs[2, 0].set_title("Normal PDF (μ=0, σ=1)")
# CDF
axs[2, 1].plot(x_norm, cdf_norm, color='red')
axs[2, 1].set_title("Normal CDF")
# Histogram of Samples
axs[2, 2].hist(samples_norm, bins=30, density=True, color='darkred', alpha=0.7
axs[2, 2].set_title("Normal Samples Histogram")

# Layout and show
for row in axs:
    for ax in row:
        ax.grid(True)
plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.show()

# ------------------- BASIC STATISTICAL ANALYSIS -------------------
print("Binomial Distribution:")
print(f"Mean = {np.mean(samples_binom):.2f}, Variance = {np.var(samples_binom)
print("\nPoisson Distribution:")
print(f"Mean = {np.mean(samples_poisson):.2f}, Variance = {np.var(samples_pois
print("\nNormal Distribution:")
```

```python
print(f"Mean = {np.mean(samples_norm):.2f}, Standard Deviation = {np.std(sampl
```




```
Binomial Distribution:
Mean = 5.00, Variance = 2.51

Poisson Distribution:
Mean = 4.07, Variance = 3.99

Normal Distribution:
Mean = 0.06, Standard Deviation = 0.98
```

Conclusion

This experiment successfully demonstrated the implementation and analysis of three key probability distributions — Binomial, Poisson, and Normal — in Python.

The Binomial distribution models discrete success/failure outcomes over fixed trials. The Poisson distribution effectively captures rare, independent events over time or space. The Normal distribution represents continuous, symmetrical data and forms the foundation of statistical analysis and machine learning.

The Central Limit Theorem highlights why the Normal distribution is universally applicable, and understanding skewness helps in selecting the right distribution for real-world data. Overall, this experiment reinforced that choosing the correct probability distribution is essential for accurate data modeling and analysis.